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Sensitivity Analysis of a TIR-Based Cold-Water Patch Detection Algorithm

The Influence of Cold Water Tributaries on Cold Water Patches

Bachelor Thesis

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Chapter 1

Introduction

In recent years, thermal infrared (TIR) remote sensing from unmanned aerial vehicles (UAVs) and other airborne platforms has become an established approach for mapping thermal heterogeneity in river systems ([Casas-Mulet et al., 2020](#); [Dugdale, 2016](#); [Kuhn et al., 2021](#); [Torgersen et al., 2001](#)). Mapping this heterogeneity is ecologically important: the cold-water patches it reveals can act as refugia that are essential for the survival of thermally sensitive fish species, particularly as river temperatures in Switzerland are expected to rise due to climate change. Despite this importance, there is currently no comprehensive information about the presence, scale, and location of such cold-water patches in Swiss water bodies ([Tonolla and Antonetti, 2021, 2026](#)).

As part of a cantonal project aiming to quantify the temperature heterogeneity and detect cold-water patches in the Emme River and its tributaries, TIR imagery covering over a 100 km river transect were collected in a helicopter-based survey. These TIR images were then processed to extract longitudinal and transversal temperature gradients, as well as to detect cold-water spots. Here, a newly developed algorithm for the automatic detection of cold-water patches is deployed with the goal of mapping these patches. Detailed information regarding the whole workflow can be found in ([Tonolla and Antonetti, 2021, 2026](#)).

The CWP detection algorithm relies on externally configured parameters, and sensitivity analyses performed in ([Tonolla and Antonetti, 2026](#)) indicate that the detection results can vary significantly based on the chosen parameter values, with the step length and the temperature delta parameters having a particularly strong influence. A systematic and more extensive sensitivity analysis of the algorithm is therefore needed to better characterise this variance and to inform potential algorithmic improvements. Performing such an analysis requires executing the algorithm across a large number of parameter combinations. The existing algorithm was found to require between 1.25 and 2 hours per run on the available hardware, rendering a full sensitivity analysis infeasible without a faster implementation. A speed-optimized reimplementation is therefore a prerequisite for the sensitivity analysis itself.

Research Questions

Based on the above, this thesis pursues three objectives: first, to investigate whether the existing CWP detection algorithm can be reimplemented in a computationally more

efficient form while producing equivalent outputs; second, to systematically investigate the algorithm’s sensitivity to its parameters, with particular focus on how the presence of tributary inflows interacts with this sensitivity; and third, to derive adjustments to the detection algorithm that could reduce this parameter sensitivity while maintaining ecologically plausible detection results. These objectives are formalised in the following three research questions:

Research question 1 Can the existing CWP detection algorithm be reimplemented in a computationally more efficient form, while producing outputs that are equivalent to those of the original implementation?

Research question 2 How does the step length parameter contribute to the variance of the detected CWP area for tributary-caused CWPs, compared to non-tributary-caused CWPs? The algorithm defines a CWP by comparing individual pixel temperatures against a local median reference temperature calculated over a defined segment whose length is governed by the step length parameter. Cold-water tributaries introduce sharp longitudinal temperature gradients which can skew this calculation if the step size is chosen sub-optimally. It is therefore hypothesised that step length sensitivity is elevated near tributary inflows, giving rise to the following hypotheses:

H_0 The sensitivity of the detected CWP area to the step length parameter does not differ between tributary-caused and non-tributary-caused CWPs.

H_1 The sensitivity of the detected CWP area to the step length parameter is significantly higher for tributary-caused CWPs than for non-tributary-caused CWPs.

Research question 3 Building on the findings from research questions 1 and 2: what adjustments could be made to the detection algorithm to reduce variability with respect to its parameters while also achieving plausible detection results?

Chapter 2

Theoretical Background

2.1 Model Input Sensitivity Analysis

The inputs to models can diver and are generally dependent in their structure and manyfoldness on the nature of the model itself. Under the light of sensitivity analysis however, it can be worthwhile to define, what the inputs to a model actually contains. In sensitivity analysis, everything that causes variation in the output of the model is classified as an input ([Saltelli et al., 2007](#)). Providing an example: If a linear regression of a target variable Y is performed, then not only are the predictors $X_1, \dots, X_n$ considered as model inputs, but also the parameters $\theta_1, \dots, \theta_n$ that correspond to the predictors. So the model:

$$Y = f(X_1, X_2, \theta_1, \theta_2) \tag{1-1}$$

would have four model inputs from the view of sensitivity analysis.

2.2 Sensitivity Analysis

According to Saltelli et al. sensitivity analysis is “The study of how uncertainty in the output of a model [...] can be apportioned to different sources of uncertainty in the model input” ([Saltelli, 2004](#)). Sensitivity analysis provides a range of benefits. Firstly it allows the modeller to apply a strategy called “Factor Fixing”: if the model output Y is found to be independent of one or multiple inputs X_i , then these can be set to fixed constants. It may then even be possible to condense a complete submodule of a model if all parameters connected to that submodule are noninfluential, making sensitivity analysis a tool for model simplification ([Saltelli et al., 2007](#)). Secondly, sensitivity analysis generally improves model validity by identifying highly influential and therefore critical model components ([Ligmann-Zielinska, 2013](#)).

2.2.1 The Morris Method

The Morris method belongs to the group of screening methods in sensitivity analysis. This can be done as these methods only require a relatively small number of model evaluations to provide sensitivity measures for the input parameters, making them a good choice for models that have many input parameters or are expensive to compute ([Campolongo et al.,](#)

2007; Saltelli et al., 2007). As a global method, Morris explores the entire input space by varying all input parameters simultaneously across their plausible ranges, meaning that sensitivity scores reflect average behaviour across the full parameter space rather than being tied to a single point (Saltelli and Annoni, 2010). The following explanation of the Morris method closely follows (Saltelli et al., 2007) and is explained more explicitly in chapter 3 of their book.

Elementary Effects and the Sampling Grid

At the core of this introduction sits a k -dimensional model Y with inputs $X_1, \dots, X_k$. These inputs can also be written in a vector $\mathbf{X}$. The model inputs are defined in the k -dimensional unit cube, meaning that X_i can take on a value in the interval $[0, 1]$. It is worth pointing out, that every model with a continuous input can be scaled to the $[0, 1]$ range, which makes the unit cube a great place to define methods. In the Morris method, this input space is discretized into a grid Ω . This grid features p levels.

An elementary effect of the i th input parameter X_i is then defined as:

$$EE_i = \frac{Y(X_1, X_2, \dots, X_{i-1}, X_i + \Delta, \dots, X_k) - Y(X_1, X_2, \dots, X_k)}{\Delta} \quad (2-2)$$

Where Δ represents some step in the unit cube. Δ can be chosen from a range of different values, which are given by the set $\left\{\frac{1}{p-1}, \dots, 1 - \frac{1}{p-1}\right\}$.

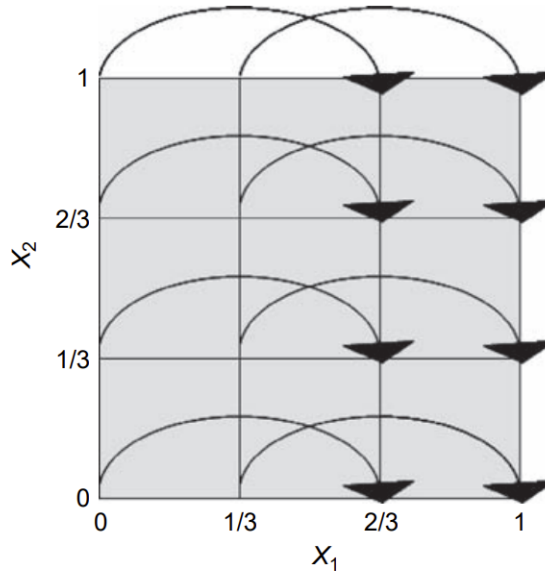


Abb. 2-1: An example grid Ω . The grid features two parameters X_1, X_2 , meaning that it is a two-dimensional grid. Further, it features four levels ($p = 4$). The step length Δ chosen here is $\frac{2}{3}$, which corresponds to an element of the set of viable values for Δ : $1 - \frac{1}{4-1} = 1 - \frac{1}{3} = \frac{2}{3}$.

The Morris method occupies itself with finding the statistical distribution of each of the i elementary effects. $\mathcal{F}_i$ is thereby used to denote the distribution of EE_i , in other words $\mathcal{F}_i \sim EE_i$ (Saltelli et al., 2007).

Trajectory-Based Sampling

To populate $\mathcal{F}_i$ with multiple samples of EE_i , the Morris method uses an efficient trajectory-based sampling design that obtains as many elementary effects as possible from as few model evaluations as possible. The exact mathematical formulation of how trajectories are constructed is beyond the scope of this thesis; a visual understanding is sufficient. Figure 2-2, taken from (Saltelli et al., 2007), shows an example trajectory through a three-dimensional sampling grid.

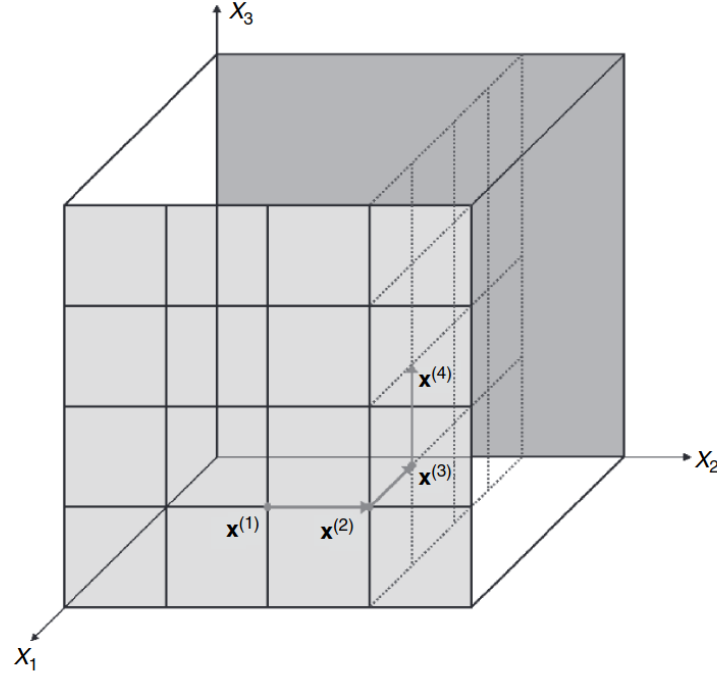


Abb. 2-2: An example Morris trajectory through a three-dimensional sampling grid ($k = 3$, $p = 6$). The trajectory consists of $k + 1 = 4$ points, with each consecutive pair of points differing in exactly one parameter. Reproduced from Saltelli et al. (2007).

From this figure it becomes visible that a trajectory consists of $k+1$ points $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(k+1)}$, where k is the dimension of the sampling grid. Each point differs from its predecessor by a step of Δ in exactly one grid direction, so that consecutive points differ in only one parameter. The trajectory is constructed such that each dimension of the grid is stepped along exactly once. Each trajectory yields exactly k elementary effects EE_i , one for each parameter spanning the sampling grid and features $k + 1$ model evaluations.

To build up $\mathcal{F}_i$ for every factor, r such trajectories are generated. Since each trajectory contributes exactly one elementary effect per factor, each $\mathcal{F}_i$ ends up containing r samples. The total number of model evaluations required is:

$$N = r \cdot (k + 1) \quad (2-3)$$

(Saltelli et al., 2007).

Choosing Δ and r

The choice of p and Δ are not independent. When p is chosen to be even, the choice $\Delta = \frac{p/2}{p-1}$ has the additional property of guaranteeing equal-probability sampling from the grid, meaning that all grid levels are sampled with equal probability across the r trajectories (Saltelli et al., 2007). There is no clear rule regarding the number of trajectories r to pick. Generally, as the number of levels p in the grid increase, the number of trajectories r also needs to increase. Otherwise it does not serve anything to increase the level of p as the grid will not sufficiently be explored to even feature all the levels when the number of trajectories is too small (Saltelli et al., 2007). Generally the number of trajectories is chosen somewhere between 10 - 50 (Campolongo et al., 2007).

Sensitivity Measures: μ , σ , and μ^*

Once the distributions $\mathcal{F}_i$ have been mapped out, statistical measures can be computed, specifically the mean μ_i and the standard deviation σ_i of $\mathcal{F}_i$ are of interest. These two statistical measures together already allow for reasonable insight into the influence of the i th parameter X_i , which corresponds to the i th elementary effect EE_i , onto the model:

If μ_i deviates strongly from 0, then this means that the elementary effect EE_i generally takes on non-zero values and therefore that the model output changes notably when a step of Δ is performed along the dimension spanned by X_i . This then ultimately means, that changes in X_i seem to generally affect the model output Y considerably and X_i therefore receives a high sensitivity score.

If σ_i is large, this means that the distribution $\mathcal{F}_i$ is spread out considerably over a larger range. This indicates, that the elementary effects EE_i that were computed were generally different depending on their location in the grid. This behaviour points either to nonlinearity regarding X_i or to the fact that X_i interacts with other input parameters. Which effect is responsible for the high standard deviation (or whether both are responsible for the effect at the same time) can not be determined. Which is a shortcoming of the method of Morris (Saltelli et al., 2007). Figure 2-3 provides an visual example of possible F distributions of a Factor X_i with the associated μ and σ values and the corresponding interpretation.

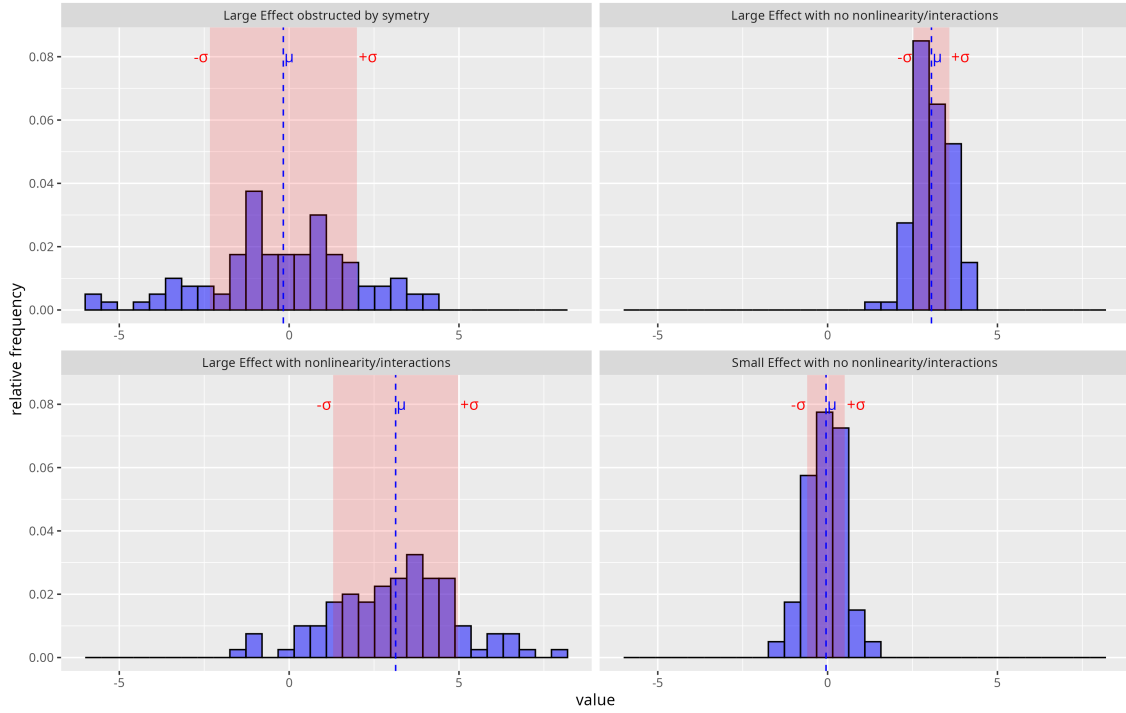


Abb. 2-3: Four example distributions $\mathcal{F}_i$ of a parameter X_i , illustrating how μ and σ jointly characterise parameter influence. *Top left:* a large effect obscured by symmetry. *Top right:* a large effect with no nonlinearity or interactions. *Bottom left:* a large effect with nonlinearity or interactions. *Bottom right:* a small effect with no nonlinearity or interactions.

Situations can occur, where $\mathcal{F}_i$ contains many extreme values however the distribution is symmetric around zero. One such example is visualized in 2-3 in the upper left plot. In such cases computing the mean value μ_i will yield a value close to zero and we would assume, that the parameter X_i therefore is noninfluential. This would be a faulty assumption, as the sampled elementary effects EE_i are substantial in their magnitude but just cancel themselves out. This phenomenon can be spotted by always taking σ_i into consideration, as in this scenario σ_i is large, but it is more common to also sample an additional distribution $\mathcal{G}_i$ which captures the magnitude of the elementary effects $|EE_i|$. So $\mathcal{G}_i \sim |EE_i|$. We then compute the mean of this distribution μ_i^* which can be interpreted like μ_i . But since it is the magnitude, the distribution never features negative values and therefore cancellation effects can not occur (Saltelli et al., 2007). Figure 2-4 provides a visual example for this phenomenon.

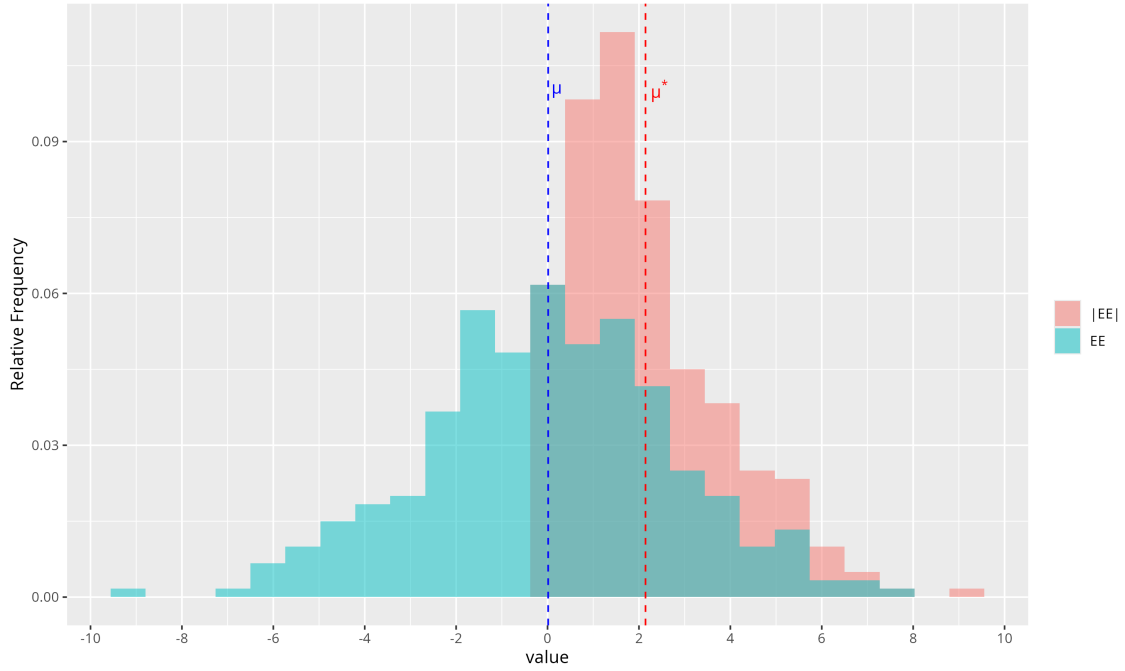


Abb. 2-4: The difference of μ and μ^* visualized with their underlying distribution $\mathcal{F}_i$ and $\mathcal{G}_i$. While the mean μ of the distribution $\mathcal{F}_i$ lies at around zero, μ^* significantly deviates from zero.

2.2.2 Application to Spatial Models

The Morris method, like most global sensitivity analysis methods, is defined for models with a scalar output: a single value Y is required for each model evaluation (Saltelli et al., 2007). The CWP detection algorithm, however, produces a spatially distributed output. Applying the Morris method to such a model therefore requires a spatial aggregation step to aggregate the spatial output to a scalar value e.g. using an area weighted mean computation Lilburne and Tarantola (2009).

2.3 Coldwater Patches

Cold water patches (CWPs) are areas in rivers and streams where the water temperature is measurably colder than that of the main channel. The temperature difference used to define a CWP varies across the literature but is typically set between 0.5 and 3 degrees Celsius below the main channel temperature (Casas-Mulet et al., 2020; Wawrzyniak et al., 2016). Cold water patches are not automatically cold water refugia. Cold water refugia are cold water patches that are actively used by cold-water organisms for thermal relief (Mejia et al., 2023). The ecological importance of cold water refugia is expected to increase as river temperatures rise due to climate change (Kurylyk et al., 2015).

Chapter 3

Methods

This chapter describes the operationalisation of the three research questions into a set of methodological steps, which are then described in detail. The first section addresses research question 1 by describing the reimplementation of the CWP detection algorithm and the comparison of its outputs and runtime against the original. The second section addresses research question 2 by describing the Morris sensitivity analysis, including the selection and classification of CWPs, the computation of sensitivity indices, and the subsequent visualisation and statistical modelling. The third section addresses research question 3 by describing the algorithm modifications derived from the sensitivity analysis findings. A description of the data and study area is provided first, as it underpins all three sections.

3.1 Data and Study Area

The Emme river originates in an extensive moorland between the Hohgant and Augstmatthorn massifs and flows through the Emmental region before joining the Aare near Zuchwil/Luterbach. The catchment covers approximately 974 km² with a mean elevation of 845 m a.s.l., and land cover is dominated by agricultural land (52.1%) and forested areas (37.9%). During summer thunderstorms, the river can rise rapidly within minutes and transport large quantities of driftwood; as a consequence, the lower reaches have been heavily engineered ([Tonolla and Antonetti, 2026](#)).

[Tonolla and Antonetti \(2026\)](#) acquired high-resolution TIR and RGB orthophotos along a 69 km transect of the Emme main channel, spanning from Kemmeribodenbad BE to Biberist SO, as well as imagery of three tributary rivers: the Ilfis, Grüene, and Urtene. The TIR imagery was collected from a helicopter platform using a FLIR SC655 thermal infrared camera. In this thesis, only the TIR and RGB orthophotos of the 69 km main channel section are used. The TIR orthophotos serve as the primary input to the CWP detection algorithms, while the RGB orthophotos are used in a downsampled state as background images for selected visualisations and were not used for any further analytical purpose. The 69 km river transect is covered by three partially overlapping single-band TIR orthophotos. Table 1-1 summarises their technical specifications.

Tab. 1-1: Technical specifications of the three TIR orthophotos covering the Emme main channel. All rasters are in the Swiss coordinate reference system CH1903+/LV95 (EPSG:2056) and stored as single-band Float32 GeoTIFF files.

	Emme Oberlauf 1	Emme Oberlauf 2	Emme Unterlauf
Extent	Kemmeribodenbad – Schüpbach	Schüpbach – Goldbach	Grüne – Biberist
Total pixels	4.75×10^9	8.45×10^8	6.57×10^9
Pixel size (m)	0.207×0.207	0.204×0.204	0.207×0.207
Min temp. (°C)	7.43	10.08	12.50
Max temp. (°C)	33.67	30.94	34.10
Mean temp. (°C)	20.85	18.84	20.90
Std dev. (°C)	1.05	1.24	1.32

3.2 Addressing Research Question 1: Algorithm Reimplementation

Performing the Morris sensitivity analysis requires $r \times (k + 1)$ model evaluations (Equation 2-3); for $k = 3$ parameters and $r = 64$ trajectories this amounts to 256 executions. Given that the existing CWP algorithm requires between 1.25 and 2 hours per run, parallel execution on ZHAW’s High Performance Cluster (HPC) is necessary to complete the experiment within a feasible timespan. To ensure that optimisation gains carry over to the HPC infrastructure, all benchmarking was performed directly on the cluster. In the remainder of this thesis, the original implementation is referred to as v1 and the speed-optimized reimplementation as v2.

3.2.1 Reimplementation of the CWP Algorithm based on GDAL and System Calls

The reworked algorithm preserves the computational logic of the original implementation but differs fundamentally in how each step is mechanistically executed. Where the original algorithm processes the raster slab-by-slab in memory using R loops and `terra/sf` functions, the reworked algorithm replaces these with calls to the GDAL command line interface, operating directly on raster and vector files on disk rather than holding intermediate results in memory. Table 2-2 gives a high-level overview of the reimplemented algorithm; the tool or function used in each step is listed in the right-hand column. The remainder of this section motivates the key design decisions underlying this choice of implementation.

Tab. 2-2: High-level overview of the reworked CWP detection algorithm, including the primary tool or function used for each step.

Step	Description	Tool / Function
1	Divide the river reach into n centerline segments; construct a wide zone polygon and a narrow reference-strip polygon for each segment.	<code>sf::st_buffer</code>
2	Rasterize the zone polygons to a raster matching the TIR raster resolution, using the segment id as burn-in value.	<code>gdal vector</code> <code>rasterize</code>
3	Round the TIR raster using a fixed rounding factor.	<code>gdal raster calc</code>
4	Extract the median temperature per segment from the rounded TIR raster within each reference strip; store in a look-up table keyed by segment id.	<code>exactextractr::exact_extract</code>
5	Reclassify the zone raster, replacing each segment id with its reference temperature from the look-up table, yielding the reference raster.	<code>gdal raster</code> <code>reclassify</code>
6	Subtract the rounded TIR raster from the reference raster; flag pixels where the difference exceeds ΔT , producing a binary CWP raster.	<code>gdal raster calc</code>
7	Polygonize the binary CWP raster into patch polygons.	<code>terra::as.polygons</code>
8	Filter patches by minimum area, compute per-patch temperature statistics, and apply a patch-level threshold check against the reference temperature.	<code>exactextractr::exact_extract</code>

Performance. The decision to heavily rely on GDAL was made as `terra` operations are sometimes not performant, and GDAL offers an often faster set of geocomputational tools, which can be called from R using `system()` or `sf::gdal_utils()` (Lovelace et al., 2019). Empirical testing in a prior coursework exercise also showed a significant speedup of GDAL-based raster operations compared to `terra` across a range of use cases (Dürig, 2026); this coursework was not peer-reviewed and is cited here only as a supporting empirical observation.

Memory footprint and HPC suitability. Operating directly on disk rather than in memory keeps the memory footprint small, which is important in an HPC setting where available memory can become a limiting factor under parallel execution. The drawback is that intermediate rasters can reach up to 25 GB in size; this was addressed by applying `COMPRESS=DEFLATE` with `PREDICTOR=2` when writing GeoTIFF files (GDAL/OGR

[contributors, 2026](#)), which substantially reduces disk space consumption with negligible computational overhead.

Raster value extraction. `terra::extract()` was replaced with `exact_extract()` from the `exactextractr` package (steps 4 and 8 of Table 2-2), which extracts aggregated raster values considerably faster for large rasters ([Mieno, 2022](#)). The two functions differ in how partially covered cells are handled: `exact_extract()` weights cells by their coverage fraction ([ISciences, LLC, 2026](#)), whereas `terra::extract()` includes a cell only if its centre point falls within the polygon ([Hijmans et al., 2024](#)) – the behaviour used in v1. To approximate this, only cells with a coverage fraction of at least 50% were included in `exact_extract()`; this approximation however is not exactly equivalent to `terra`’s centre-point rule.

3.2.2 Output Equivalence and Runtime Comparison

The resulting reimplementaion was tested for equivalence in output by comparing the CWP polygons produced by v2 against those produced by v1 when deployed with the exact same set of parameters. Spatial agreement between the two sets of polygons was assessed using the Jaccard similarity index:

$$J(A, B) = \frac{\text{area}(A \cap B)}{\text{area}(A \cup B)} \quad (2-1)$$

where A and B denote the sets of CWP polygons produced by v1 and v2 respectively, $\text{area}(A \cap B)$ is the total area of spatial overlap between the two polygon sets, and $\text{area}(A \cup B)$ is the total area covered by at least one of the two polygon sets. A value of $J = 1$ indicates perfect spatial agreement, while $J = 0$ indicates no overlap. In addition to this quantitative measure, visual inspection of the polygon outputs in QGIS was carried out to identify any systematic spatial patterns in the disagreements. Table 2-3 shows the parameter combinations used in testing.

Further, the runtime of v2 was compared against that of v1 with respect to the step length parameter. Runtime was compared across three segment lengths (400, 800, and 1200 m), with all other parameters held constant (zone halfwidth = 60 m, buffer = 3 px, $\Delta T = 1.0^\circ\text{C}$, minimum patch area = 2 m^2 , rounding = 0.1°C , 8-connectivity enabled).

Tab. 2-3: Parameter combinations used for the output equivalence comparison between v1 and v2.

Parameter	Test 1	Test 2	Test 3
segment_length_m	200	800	1200
zone_halfwidth_m	60	60	60
buffer_px	4	3	5
delta_C	0.5	1.0	1.5
min_patch_area_m2	2	3	2
round_to	0.1	0.1	0.1
connect_diagonals	TRUE	TRUE	TRUE

3.3 Addressing Research Question 2: Sensitivity Analysis

With the algorithm optimized, sensitivity analysis can now be carried out to determine the sensitivity of the CWP detection algorithm with respect to the parameters buffer pixel count, step length, and temperature delta.

The first step focuses on determining CWP that are susceptible to the step length parameter and classifying these CWP into tributary-caused and non-tributary-caused CWP. The second step derives Morris sensitivity indices for these CWP with respect to their area. The third step displays the sensitivity analysis results visually and fits a statistical model relating the sensitivity of CWP area to the step length parameter to the CWP's area and its temperature difference to the surrounding environment.

The inclusion of buffer pixel count and temperature delta alongside step length also serves a secondary, validatory purpose. [Tonolla and Antonetti \(2026\)](#) already found in their sensitivity analysis that the buffer pixel count seems to have little influence on the CWP area; if this finding can be replicated here, algorithmic simplification using factor fixing might be possible. Conversely, the temperature delta parameter was among the most influential parameters in [Tonolla and Antonetti \(2026\)](#). Replicating both findings within the present methodology would lend additional credibility to both approaches.

3.3.1 Determining and Classifying Step Length Sensitive CWP

Especially of interest is to observe the difference in sensitivity towards the step length parameter between tributary-caused and non-tributary-caused CWP. For this reason, a pre-selection method is deployed to guarantee that the studied CWP are sensitive towards the step length parameter. A Morris design with $r = 44$ and $k = 2$ is first set up for the buffer pixel count and step length parameters. Then the speed-optimized (v2) CWP algorithm is deployed on the resulting parameter tuples. Lastly, the results are aggregated to a fuzzy representation of the CWP over all runs, to then select CWP which are variable with respect to the CWP algorithm and to classify them into tributary-caused and non-tributary-caused CWP. Figure 3-1 provides a visual overview of these steps.

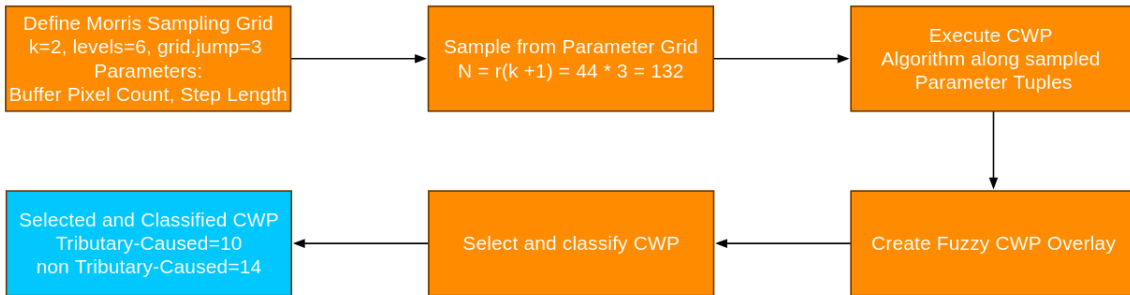


Abb. 3-1: Visual overview of the steps involved in determining and classifying step-length-sensitive CWP: setting up the first Morris experiment and executing the CWP algorithm across the sampled parameter tuples, aggregating the resulting polygon layers into a fuzzy overlay raster, and selecting and classifying variable CWPs into tributary-caused and non-tributary-caused groups.

Setting up First Morris Experiment

First, a two-dimensional Morris sampling grid was set up spanning two parameters: buffer pixel count and step length. A total of 6 levels were set for each parameter, with the step length varying from 400 m to 1200 m and the number of pixels in the buffer spanning from 2 to 7. To compute the elementary effects, a jump distance of 3 was set. Figure 3-2 shows the deployed grid visualized in greater detail, with the jump distance additionally drawn in.

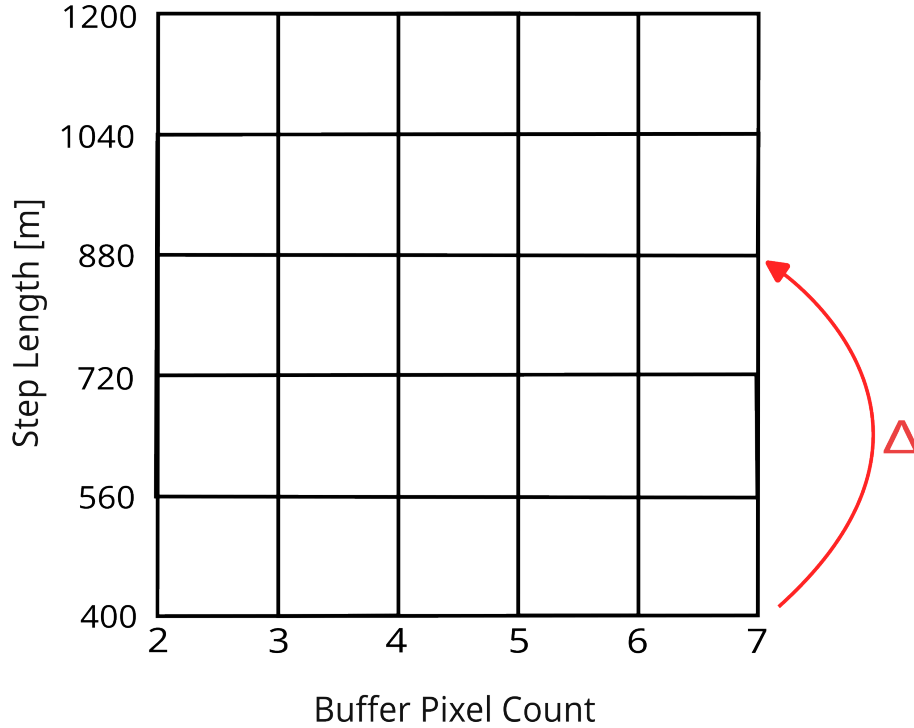


Abb. 3-2: Visual depiction of the Morris sampling grid, spanning step length and buffer pixel count. The arrow shows the Δ chosen for the Morris experiment, corresponding to a jump of 3 grid cells.

From this grid, a total of 44 trajectories were sampled ($r = 44$), resulting in a total of $44 \times (2 + 1) = 132$ parameter tuples being generated. Each tuple holds a different combination of buffer pixel count and step length. These were written to a file along with a unique identifier for each tuple. For the sampling procedure, the R package **sensitivity** was used.

The decision to vary only the buffer pixel count and step length in this first Morris experiment (i.e., $k = 2$, with the temperature delta parameter held constant) is methodologically important. The temperature delta parameter directly determines the threshold used by the algorithm to flag pixels as part of a CWP; varying it would therefore be expected to introduce variability in the detected extent of essentially all CWPs, independent of whether they are sensitive to the step length parameter specifically. Including the temperature delta in this first step would consequently make it difficult to distinguish CWPs that are variable because of the step length parameter from CWPs that merely appear variable because the CWP-defining threshold itself was varied. By holding the temperature delta constant and varying only step length and buffer pixel count, any variability observed in the resulting fuzzy overlay can be more confidently attributed to these two parameters, and in particular to step length (as the buffer pixel count is generally non-influential).

A Morris design was used for this first step, rather than e.g. a simple grid sweep, to ensure the validity of the selection criterion itself: the goal of this step is to select CWPs whose variability can subsequently be quantified using the Morris sensitivity indices computed in the second experiment. Had a different sampling scheme been used here, it would be possible to visually identify CWPs as variable based on a pattern of variation that the Morris elementary effects might not register as sensitive. By already constructing the fuzzy overlay from Morris-sampled runs in this first step, any variability visible in the overlay is, by construction, of the kind that the subsequent Morris-based sensitivity analysis is designed to detect.

Model Evaluation along Parameter Combinations

Next, a parallelized computation scheme was set up to evaluate the CWP algorithm along the created parameter tuples, while keeping the parameters not part of the Morris experiment constant. Table 3-4 shows the values at which the other parameters were held constant.

Tab. 3-4: Parameter values held constant during the first Morris experiment, while the step length and the buffer pixel count were varied according to the sampled parameter tuples.

Parameter	Value
zone_halfwidth_m	60 m
delta_C	1.0 °C
min_patch_area_m2	2 m ²
connect_diagonals	TRUE
round_to	0.1

To orchestrate the parallel execution of the CWP algorithm, the open source software GNU Parallel was deployed (Tange, 2024). Specifically, GNU Parallel was used to implement a worker pool of processes, each executing the CWP algorithm with a parameter tuple fetched from a parameter text file. Before starting the computation, each process also sets up an individual working directory, to which intermediate results as well as the final result are written. To keep the parameter combination as well as the identifier traceable – and the results therefore identifiable – the folder name holds both the identifier and the parameter values. Figure ?? visualizes the GNU Parallel workflow.

Creation of a Fuzzy Overlay of CWP

The resulting CWP polygon layers were then processed into a single raster through a series of processing steps. First, each polygon layer was rasterized using the same pixel size as the TIR rasters. Then, these different raster representations of the CWPs were stacked into a multi-layer raster, and a pixel-wise mean calculation was performed. This aggregates the three-dimensional raster stack into a two-dimensional raster. This process is visualized in Figure 3-3.

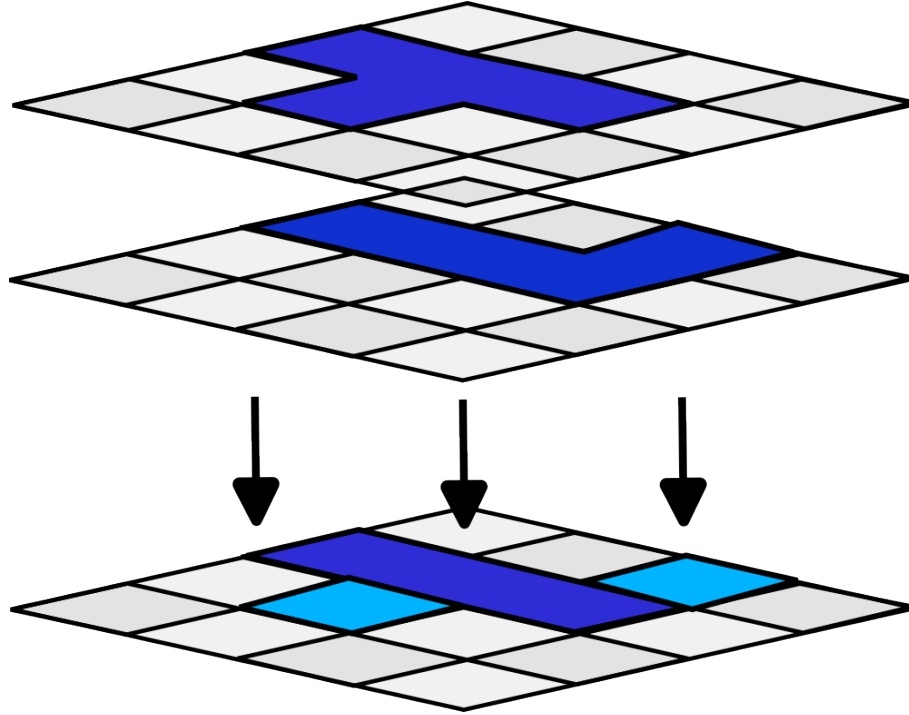


Abb. 3-3: Illustration of the fuzzy overlay construction. The two upper rasters show the binary CWP detection outputs of two individual Morris runs, where dark blue pixels indicate detected CWP and white pixels indicate non-CWP area. The lower raster shows the resulting fuzzy overlay, where light blue pixels hold a lower score than dark blue pixels, reflecting locations where CWP detection was inconsistent across runs.

The pixel-wise mean computation produces a pixel score between 0 and 1, which can be interpreted as follows: a pixel value of 1 indicates that this area of the Emme basin was part of a CWP during each of the runs, while a pixel value near zero indicates that the area covered by the pixel was almost never part of a CWP in the model outputs. The resulting raster can be understood as a fuzzy membership representation of the CWPs in the Emme, where each pixel has a probability-like score of belonging to a CWP. From these fuzzy CWPs, it can also be read which CWPs are variable with respect to the step length parameter. CWPs which show a wide range of different shades in their coloring contain pixels that are generally variable in their identity as CWP or non-CWP pixels. These CWP are therefore candidates for the selection and classification process described in the following subsection.

Selection and Classification of CWP

From this fuzzy CWP raster, CWP which were clearly variable and whose origin could be clearly attributed to either a tributary inflow or the absence thereof were selected and classified into the groups tributary-caused and non-tributary-caused. The selection and classification process was not carried out algorithmically due to time constraints, but was based on a visual interpretation of the patches. First, patches were judged as being variable or non-variable based on their coloring in the fuzzy overlay. Patches were generally determined to be variable when their coloring contained a substantial proportion of grey and black pixels, indicating that the detected CWP extent changed considerably across runs. These CWP then underwent classification. A CWP was classified as tributary-caused if

it was located in close proximity to a tributary inflow – either directly at, or immediately downstream of, the point where the tributary enters the receiving river. Figure 3-4 visualizes this decision process as a decision tree.

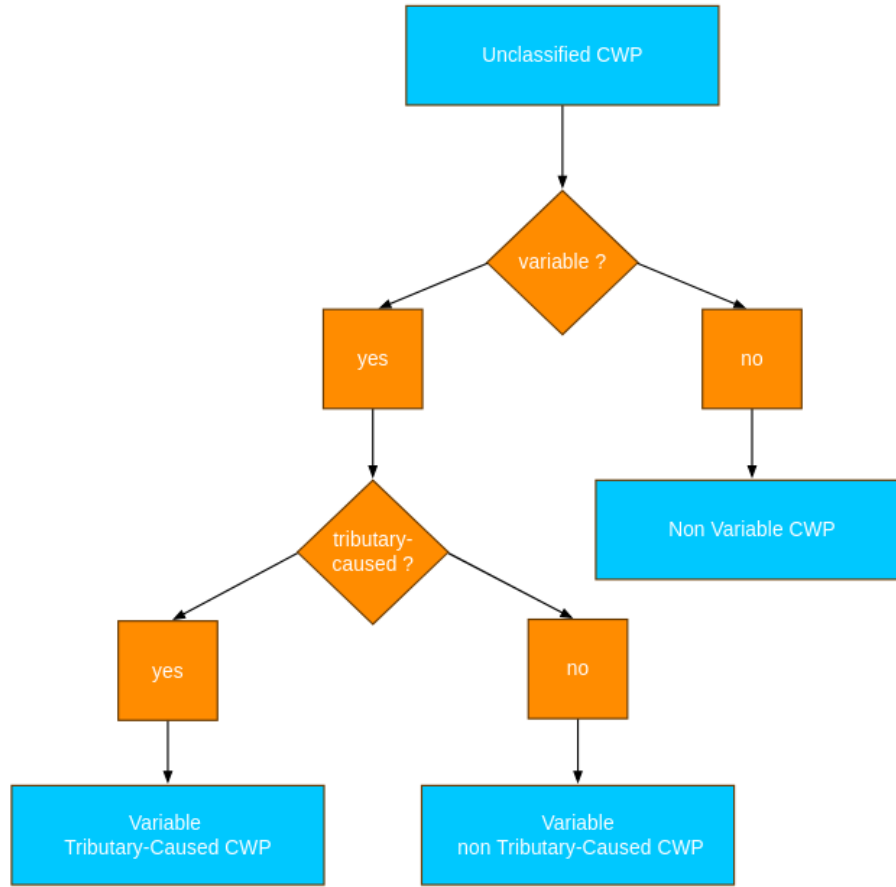


Abb. 3-4: Decision tree illustrating the visual classification process applied to candidate CWP patches: patches are first assessed as variable or non-variable based on their coloring in the fuzzy overlay, and variable patches are further classified as tributary-caused or non-tributary-caused based on their proximity to a tributary inflow.

To identify tributary inflows, OpenStreetMap as well as SwissImage were consulted. OpenStreetMap facilitated the identification of tributaries which might be obscured by forest or buildings in the SwissImage data, while SwissImage provided the opportunity to spot tributaries which might not be mapped in OpenStreetMap, as OpenStreetMap is based on volunteered geographic content and therefore cannot be assumed to be complete.

A total of 24 CWP patches which are variable and can be clearly attributed to being either tributary-caused or non-tributary-caused were selected and classified. In the context of research question 2, each of these 24 CWP and its spatial extent constitutes one of the annotated locations across which sensitivity indices are computed. For each selected CWP, a bounding box was extracted.

Multiple small adjacent CWPs were grouped together and treated as a single CWP patch. This was done with spatial autocorrelation in mind: Tobler’s first law of geography states that all things are related, but close things are more related than distant things. Small CWP located close together near the same location share the same local generating conditions (same tributary, same reach geometry, same local thermal regime). Their sensitivity

to the step length parameter will therefore be highly similar precisely because of their spatial proximity. Treating them as separate CWP would consequently over-represent that location in the analysis, as if the same spot were sampled multiple times and counted as independent observations, skewing the analysis towards those local conditions.

3.3.2 Computing Morris Sensitivity Indices with respect to CWP Area

With the 24 CWPs selected and classified, the second stage of the sensitivity analysis focuses on quantifying their sensitivity to the three parameters buffer pixel count, step length, and temperature delta. As discussed in Section ??, applying the Morris method to a spatially distributed model output requires aggregating that output into a scalar statistic per model run. Here, the total detected CWP area serves this role. The computation proceeds through three main steps: setting up and executing a second Morris experiment across the classified CWP locations, computing lumped area statistics per location and run, and deriving and analysing the Morris sensitivity indices. Figure 3-5 provides a visual overview of these steps.

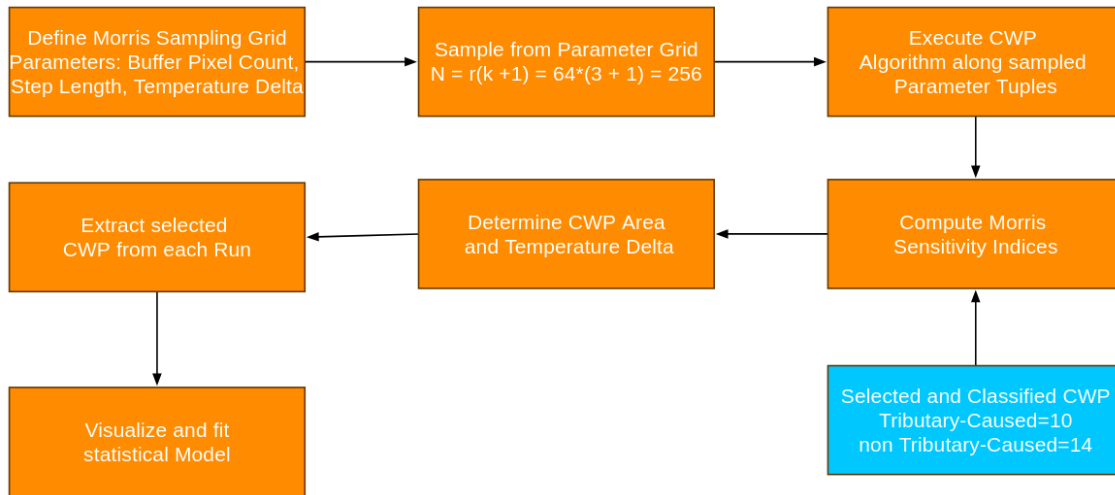


Abb. 3-5: Visual overview of the steps involved in computing Morris sensitivity indices with respect to CWP area: setting up the second Morris experiment ($k = 3$) and executing the algorithm across the sampled parameter tuples, computing lumped area statistics per annotated CWP location and Morris run, and computing and visualising the Morris sensitivity indices.

Setting up Second Sampling Grid and Executing Algorithm

A three-dimensional Morris sampling grid was set up spanning buffer pixel count, step length, and temperature delta. A visual depiction of the grid is provided in Figure 3-6.

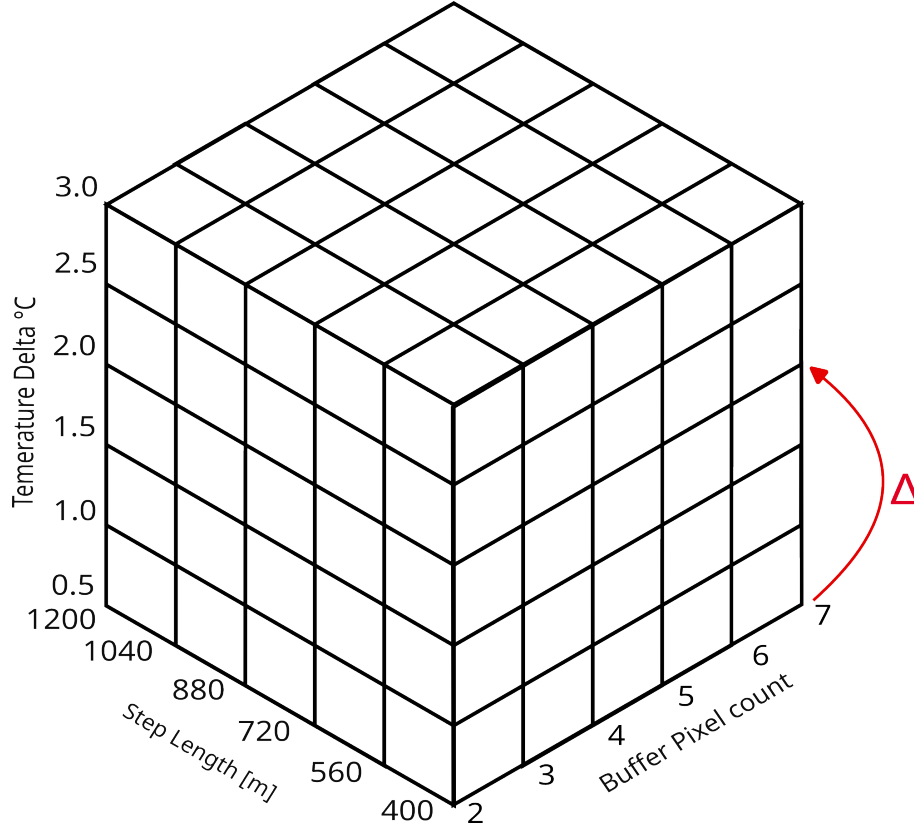


Abb. 3-6: Visual depiction of the second Morris sampling grid, spanning step length, buffer pixel count, and the temperature delta threshold. The red arrow indicates the Δ chosen.

A total of 64 trajectories were sampled from this grid, creating a total of $64 \times (3 + 1) = 256$ sampling tuples (see Equation 2-3). The model was again executed in parallel on the different parameter tuples, with the remaining parameters held constant at the values provided in Table 3-5. For the execution orchestration, the same set-up as in Section 3.3.1 was deployed. The resulting CWP polygons were again written to folders uniquely identified by an id as well as the parameter values.

Parameter	Value
min_patch_area_m2	2 m ²
zone_halfwidth_m	60 m
connect_diagonals	TRUE

Tab. 3-5: Parameter values held constant during the second Morris experiment, while `step_length`, `buffer_px`, and `delta_T` were varied according to the sampled parameter tuples.

Computing Lumped Statistics for the Selected CWPs

In the next step, the bounding boxes of the 24 selected CWPs and the polygon outputs from the second Morris experiment are brought together. For each bounding box and each Morris run, the CWP polygons lying inside the bounding box are extracted. The total area covered by CWP within the bounding box and the area-weighted mean temperature delta across all CWP polygons inside it are then computed. Figure 3-7 summarizes this

per-location, per-run computation schematically.

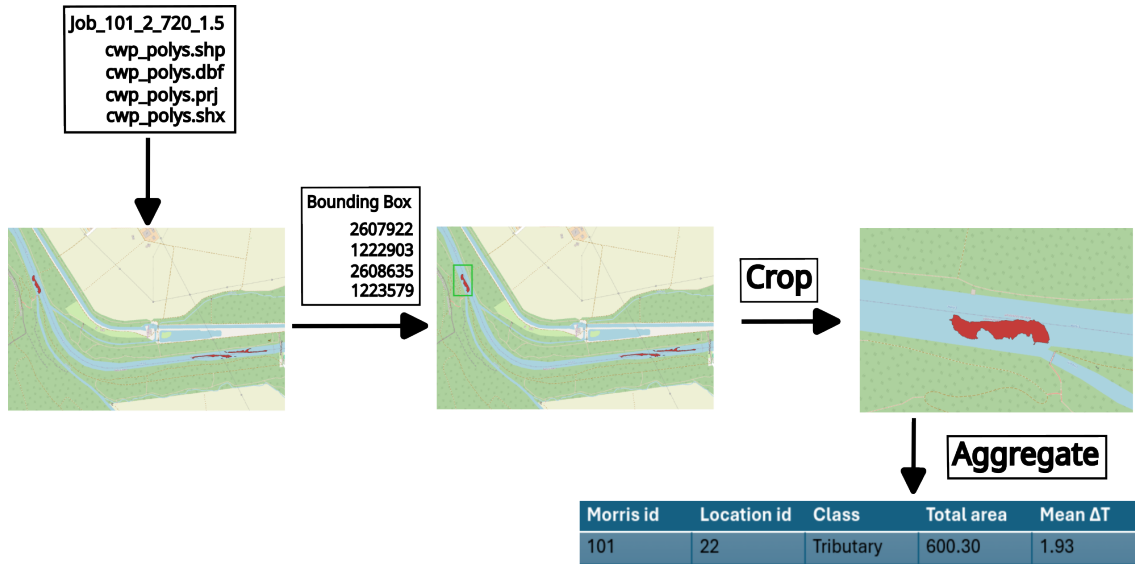


Abb. 3-7: Schematic overview of the lumped-statistics computation for one annotated CWP location (Limbach inflow into the Emme near Utzenstorf BE). For each Morris run, CWP polygons are cropped to the bounding box; if polygons are present, total area and area-weighted mean temperature delta are computed; if the bounding box is empty, both are set to zero.

Two special cases arise in this computation. First, if a bounding box contains only a single CWP polygon, the area-weighted mean temperature delta simply corresponds to the temperature delta of that polygon. Second, if a bounding box is completely empty for a given run – which occurs when the algorithm produces no CWP at that location for the corresponding parameter combination – both the total area and the temperature delta are set to zero. Setting the area to zero is straightforward: no detected CWP is equivalent to a CWP with zero surface area. Setting the temperature delta to zero is equally justified: a CWP is defined by a temperature difference to the surrounding river exceeding a given threshold; a location with no detected CWP is therefore treated as having no thermal anomaly, making a temperature delta of zero the logical neutral value.

The result of this process is a table in which each row corresponds to one annotated CWP location under one Morris parameter tuple, with the total detected CWP area and mean temperature delta as the two output measures. Table 3-6 shows an excerpt of this table.

As the table illustrates, the area measures generally differ for the same CWP location across different parameter tuples. The total CWP area is therefore a suitable scalar statistic for characterising the variability of each CWP with respect to the input parameters.

Computing Morris Sensitivity Indices

The Morris sensitivity indices μ , μ^* , and σ are then computed with respect to the total CWP area for each bounding box. Before computation, the total CWP area values are normalised per location using min-max scaling into the $[0, 1]$ range. This is necessary because without normalisation, locations with large CWP would produce larger absolute area differences than locations with small CWP, simply due to their size – making the resulting sensitivity indices incomparable across locations. After scaling, an area value of

Location	Class	ΔT [°C]	Total CWP area [m ²]
2	Non-tributary	3.67	32.32
2	Non-tributary	4.13	22.65
2	Non-tributary	4.28	18.92
⋮	⋮	⋮	⋮
4	Tributary	2.40	39.82
4	Tributary	2.91	15.89
4	Tributary	0	0
⋮	⋮	⋮	⋮

Tab. 3-6: Excerpt of the lumped statistics table. Each row corresponds to one annotated CWP location under one Morris parameter tuple. The area measures differ across rows sharing the same location, reflecting the variability of the detected CWP extent with respect to the input parameters.

1 corresponds to the largest area ever produced for that bounding box across all runs, and a value of 0 corresponds to an empty bounding box.

The sensitivity indices are then computed with respect to these normalised area values using the `sensitivity` package.

Visualisation and Statistical Modelling

Two sets of boxplot visualisations were produced. The first displays the distributions of μ^* and σ with respect to the step length parameter, stratified by CWP class (tributary-caused and non-tributary-caused), to allow a direct visual comparison of sensitivity between the two groups. The second displays the μ^* distribution with respect to all three varied parameters – buffer pixel count, step length, and temperature delta – pooled across both classes, to provide an overview of the relative influence of each parameter on CWP area.

Additionally, a linear model was fitted to gain insight into whether the mean area of a CWP and its mean temperature delta to the surrounding environment are related to its sensitivity to the step length parameter. The response variable was μ^* with respect to step length, and the two predictors were the mean total CWP area across all Morris runs for that bounding box and the mean temperature delta across all Morris runs for that bounding box. The model was fitted using R’s `lm()` function across all 24 annotated CWP locations, pooling both classes:

$$\mu_{\text{step length}}^* = \beta_0 + \beta_1 \cdot \overline{\text{area}} + \beta_2 \cdot \overline{\Delta T} + \varepsilon \quad (3-2)$$

3.4 Addressing Research Question 3: Algorithm Modification

With the 24 CWPs selected and classified, the second stage of the sensitivity analysis focuses on quantifying their sensitivity to the three parameters buffer pixel count, step length, and temperature delta. This is achieved through three steps: first, a second Morris experiment is designed and executed across the classified CWP locations; second,

the polygon outputs are reduced to a scalar statistic per location and run; and third, the Morris sensitivity indices are computed from these scalars and analysed statistically. Figure 3-5 provides a visual overview of these steps.

3.4.1 Motivation for Algorithm Modification

The sensitivity analysis of [Tonolla and Antonetti \(2026\)](#) provides direct motivation for the design of v3. Three findings from that work informed specific algorithmic decisions.

First, [Tonolla and Antonetti \(2026\)](#) found that the buffer pixel count has little influence on the detected CWP area. This suggests that the buffer pixel count qualifies as a candidate for factor fixing – that is, it can be set to a fixed constant and removed as a free parameter without meaningfully affecting the algorithm’s output. In v3, the buffer pixel count is therefore internalized and set to a constant value of 3, which lies within the range of values examined by [Tonolla and Antonetti \(2026\)](#).

Second, [Tonolla and Antonetti \(2026\)](#) identified the step length parameter as having a non-negligible influence on the detected CWP area. A plausible mechanistic explanation for this sensitivity is that CWPs spanning a slab boundary are compared against two different reference temperatures – one per slab – which can produce artificial truncations in the detected patch extent. This motivated a fundamental redesign of the reference temperature computation in v3, replacing the slab-based approach with a spatially continuous IDW-based scheme that does not depend on a step length parameter at all.

Third, [Tonolla and Antonetti \(2026\)](#) found the temperature delta parameter to be the most influential of the parameters examined. Unlike the buffer pixel count and step length, however, the temperature delta cannot simply be removed: different thresholds are used in the literature to define CWPs ([Casas-Mulet et al., 2020](#)), and the user must therefore retain the ability to set it according to their chosen definition. The temperature delta is accordingly retained as a free parameter in v3.

Whether the design decisions described above are supported by the Morris screening analysis conducted in this work will be examined in the Results and Discussion.

3.4.2 Operationalisation of Surrounding Stream Temperature

The step-length dependency was addressed by rethinking the operationalisation of the surrounding stream temperature used as the reference for pixel flagging. In v1 and v2, the reference temperature for each pixel is determined by the median temperature of the slab segment to which that pixel belongs. This approach has two consequences: it requires a user-defined step length, and it produces a reference temperature field that is piecewise constant – uniform within each slab and discontinuous at slab boundaries. CWPs that happen to span a slab boundary are therefore compared against two different reference temperatures, which can produce artificial cutoffs in the detected patch extent.

In v3, this is replaced by an inverse distance weighting (IDW) scheme. Rather than dividing the reach into fixed slabs, reference temperature sample points are obtained by sampling multiple step length values from a uniform distribution and extracting the median temperature at each resulting segment’s visual center. These sample points, pooled across all sampled step lengths, are then used to compute a spatially continuous reference temperature raster via IDW, where each cell’s reference temperature is a distance-weighted

average of the surrounding sample point temperatures. This guarantees that the reference temperature is a spatially continuous field, and that no sharp cutoffs at slab boundaries can occur.

3.4.3 Implementation Detail

The implementation of v3 follows the structure of v2 closely, differing only in the computation of the reference temperature raster (steps 1–5 below, replacing steps 1–5 of Table 2-2). The full reference temperature computation proceeds as follows:

Tab. 4-7: High-level overview of the v3 CWP detection algorithm. Steps 1–5 replace the corresponding steps of Table 2-2; steps 6–9 are identical to v2.

Step	Description
1	Sample 20 candidate step length values from a uniform distribution with bounds 500–3000 metres.
2	For each candidate step length, divide the river centerline into n segments; construct a narrow 3-pixel-wide reference strip polygon for each segment; extract the median temperature of each segment from the TIR raster; compute the visual centre of each strip polygon and assign the corresponding median temperature to this point.
3	Pool the reference points (location + temperature) from all 20 step length values into a single combined set of IDW sample points.
4	Initialise a new raster with the same extent and resolution as the TIR raster. For each cell: if the corresponding TIR cell is NaN, set to NaN; otherwise compute a distance-weighted reference temperature via IDW over the pooled sample points, using a decay power of 2.
5	Round the TIR raster using a fixed rounding factor.
6	Subtract the rounded TIR raster from the reference raster; flag pixels where the difference exceeds ΔT , producing a binary CWP raster.
7	Polygonize the binary CWP raster into patch polygons.
8	Filter patches by minimum area; compute per-patch temperature statistics; apply a patch-level threshold check against the reference temperature.

Step 4 is implemented in C++ and imported into R via the **Rcpp** package, as the per-pixel IDW computation was identified as the primary computational bottleneck.

3.4.4 Reduction of Parameters

The modifications described above result in a reduction of the algorithm’s free parameters. The buffer pixel count is fixed internally at a value of 3. The step length parameter is no longer required, as the IDW reference temperature computation samples reference points across a range of step lengths rather than relying on a single user-defined value. The zone halfwidth parameter is likewise no longer needed, as the extent of the reference raster

is matched to the extent of valid pixels in the TIR raster. It should be noted that this implies a constraint on the input data: v3 can only be applied to TIR rasters that contain an exact water mask of the river channel, with non-river pixels set to NaN.

The remaining free parameters of v3 are: the paths to the TIR raster and the river centerline, the minimum patch area, the rounding factor, the temperature delta threshold, and whether diagonal pixels are counted as neighbours during polygonization.

3.5 Usage of AI

The large language models Claude Sonnet 4.6 and Claude Opus 4.6 (Anthropic) were used as aids to ensure proper formulation, spelling, and grammatical correctness. They were additionally used to derive L^AT_EX for figures and the simulation protocol table, and as coding assistants.

Chapter 4

Results

4.1 Research Question 1: Algorithm Reimplementation

This section presents the results of the runtime benchmarking and output equivalence comparison between v1 and v2. The two subsections address the speed improvement achieved by the reimplementation and the degree to which the two implementations produce spatially consistent CWP detections.

4.1.1 Runtime Comparison

Figure 1-1 shows the execution time of v1 and v2 as a function of segment length. The execution time of v2 is approximately 0.5 hours across all three tested segment lengths, remaining largely constant. In contrast, v1 shows execution times ranging from approximately 2 hours at a segment length of 400 m to approximately 1.25 hours at 1200 m, with execution time increasing as segment length decreases (i.e., as the number of segments increases).

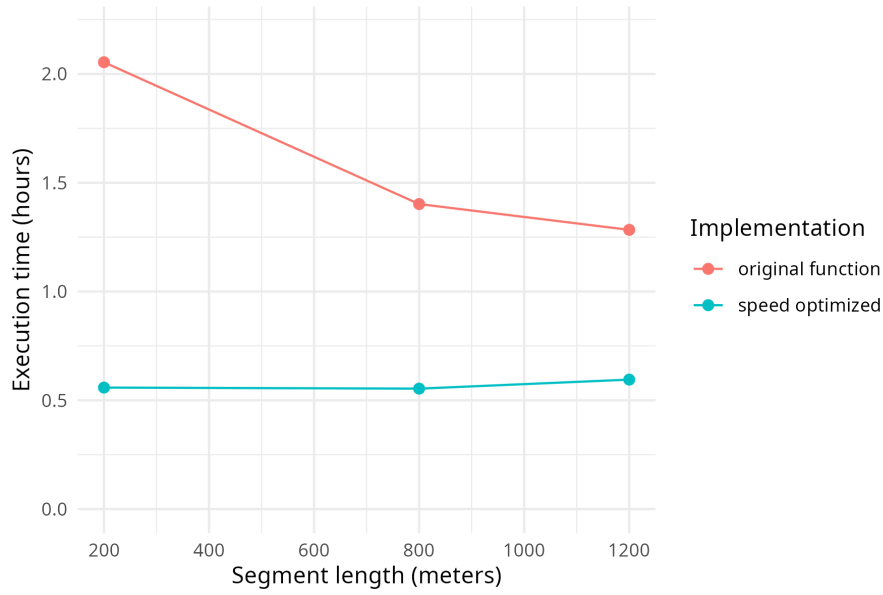


Abb. 1-1: Execution time of the original (v1) and optimized (v2) implementations of `detect_cwp_single` as a function of segment length. The optimized implementation shows execution times that are both shorter and largely constant across segment lengths, whereas the original implementation scales disproportionately with decreasing segment length (i.e., increasing number of segments).

4.1.2 Output Equivalence

The three comparison runs achieved outputs that are in general agreement with respect to CWP location and shape, but are not identical at the pixel level. The Jaccard similarities for the three runs are 0.936, 0.956, and 0.976, indicating generally good agreement between v1 and v2. In all figures, v2 polygons are shown in red and v1 polygons in blue, overlaid at 50% transparency; areas of perfect agreement therefore appear as a blend of both colours.

Figure 1-2 illustrates the range of agreement observed across the three trial runs. Panel (a) shows a representative example of the general agreement between v1 and v2 for trial run 3: the two algorithms detect the same CWPs at the same locations, with discrepancies confined to a small number of boundary pixels. Panel (b) shows a more pronounced example of pixel-level disagreement from trial run 1, where the boundary mismatch between the two polygon sets is more visible. Panel (c) shows an example of complete spatial agreement from trial run 2, where the v1 and v2 polygons are indistinguishable.

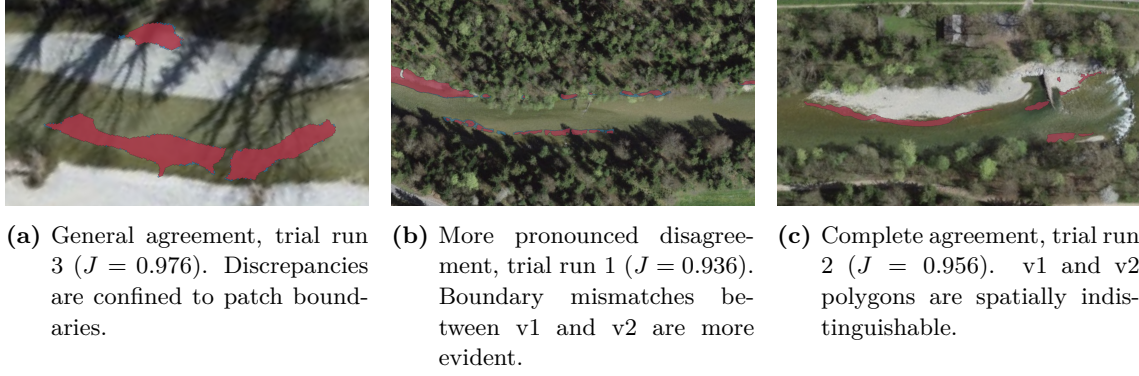


Abb. 1-2: Comparison of CWP polygon outputs between v1 (blue) and v2 (red) across the three validation runs. Polygons are overlaid at 50% transparency; areas of agreement appear as a blend of both colours, while areas covered by only one algorithm appear in that algorithm's colour alone.

4.2 Research Question 2: Sensitivity Analysis

This section presents the Morris sensitivity analysis results across three complementary views: the sensitivity of detected CWP area to the step length parameter stratified by CWP class, the overall sensitivity ranking across all three varied parameters, and the relationship between CWP area and step-length sensitivity as quantified by the linear model.

4.2.1 Sensitivity of CWP Area to Step Length

Figure 2-3 shows the Morris sensitivity indices μ^* and σ with respect to the step length parameter, stratified by CWP class ($n = 10$ tributary-caused, $n = 14$ non-tributary-caused). All non-tributary-caused CWP show μ^* values below 0.2, indicating low sensitivity to the step length parameter across this class. The tributary-caused CWP show a wider spread in μ^* : some tributary-caused CWP have sensitivity values comparable to the non-tributary-caused group, while others show substantially higher values of up to 0.3–0.6. The median μ^* values of the two classes are similar, with the primary difference lying in the upper tail of the tributary-caused distribution. The σ values show a comparable pattern, with greater spread in the tributary-caused group.

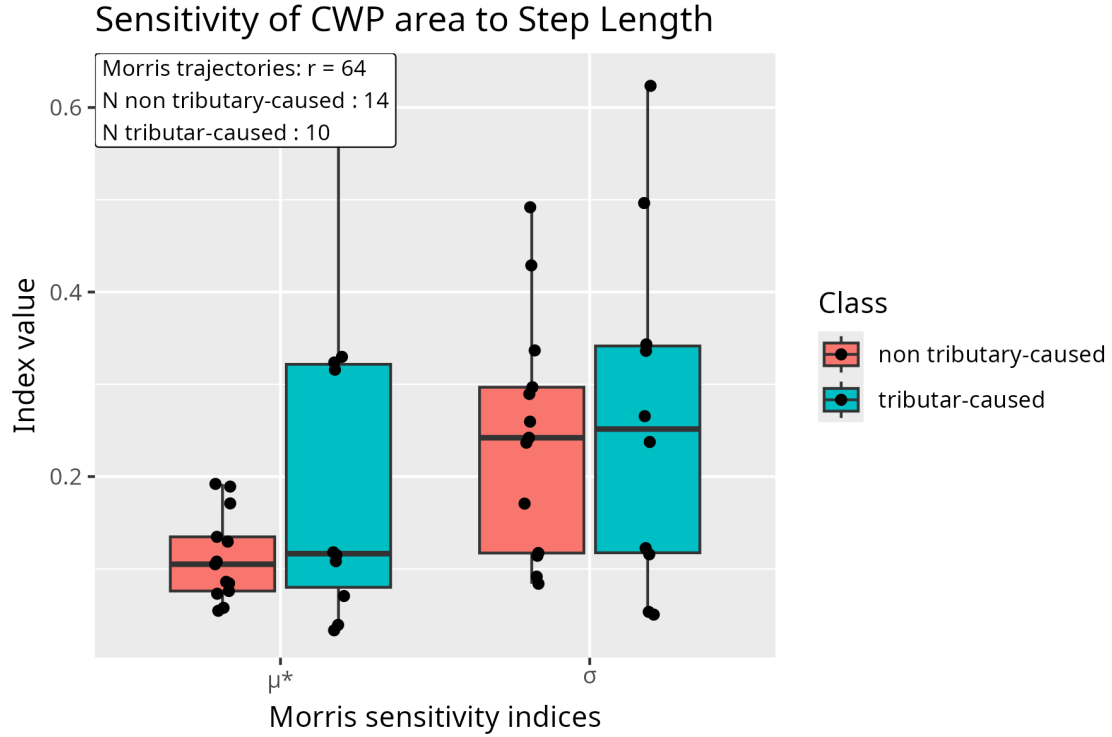


Abb. 2-3: Morris sensitivity indices (μ^* and σ) for the `step_length` parameter with respect to cold water patch area, stratified by patch class (red: tributary-caused, $n = 10$; blue: non-tributary-caused, $n = 14$). Boxplots show the distribution across bounding boxes.

4.2.2 Overall Sensitivity across all Parameters

Figure 2-4 displays the μ^* values for all 24 CWP locations with respect to the three parameters varied in the second Morris experiment. The buffer pixel count shows consistently low μ^* values across all locations, with the entire distribution concentrated near zero. In contrast, both step length and temperature delta show distributions that deviate clearly from zero. The temperature delta parameter shows the highest μ^* values overall, suggesting it has the strongest influence on the detected CWP area across the set of annotated locations.

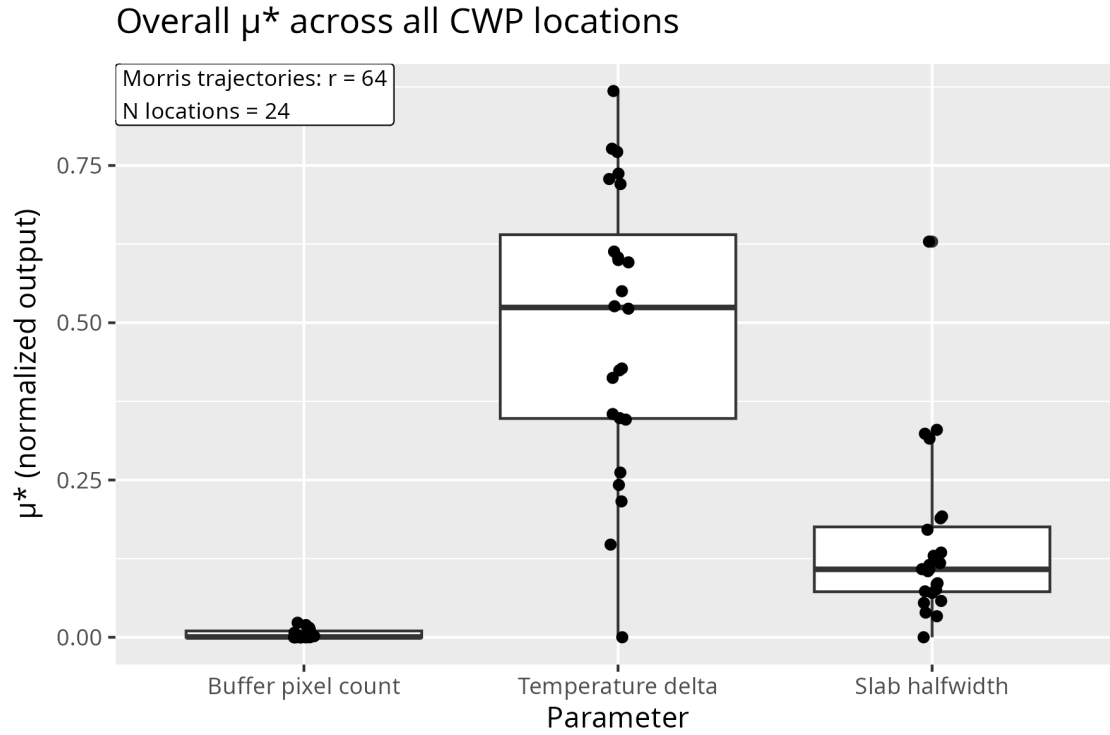


Abb. 2-4: Overall μ^* sensitivity indices for `buffer_px`, `delta_T`, and `step_length` across all classified cold water patches ($N = 24$ locations, $r = 64$ trajectories).

4.2.3 Relationship between CWP Area and Step Length Sensitivity

Figure 2-5 shows the 24 selected CWP along three dimensions simultaneously: mean CWP area averaged across all Morris runs (x-axis), μ^* with respect to the step length parameter (y-axis), mean temperature delta to the surrounding environment averaged across all Morris runs (color), and CWP class (shape). The data suggest a positive relationship between mean CWP area and sensitivity to the step length parameter: locations with larger mean CWP area tend to show higher μ^* values. This pattern appears across both classes. No clear relationship between mean temperature delta and μ^* is visible from the coloring.

Table 2-1 shows the results of the linear model fitted to predict μ^* (step length) from mean CWP area and mean temperature delta. The mean area predictor is highly significant ($p < 0.001$) with a positive slope, consistent with the visual pattern in Figure 2-5. The mean temperature delta predictor is not significant ($p = 0.556$), indicating no detectable linear relationship between temperature delta and sensitivity to the step length parameter after accounting for area. The model explains a substantial proportion of variance ($R^2 = 0.611$, Adjusted $R^2 = 0.572$).

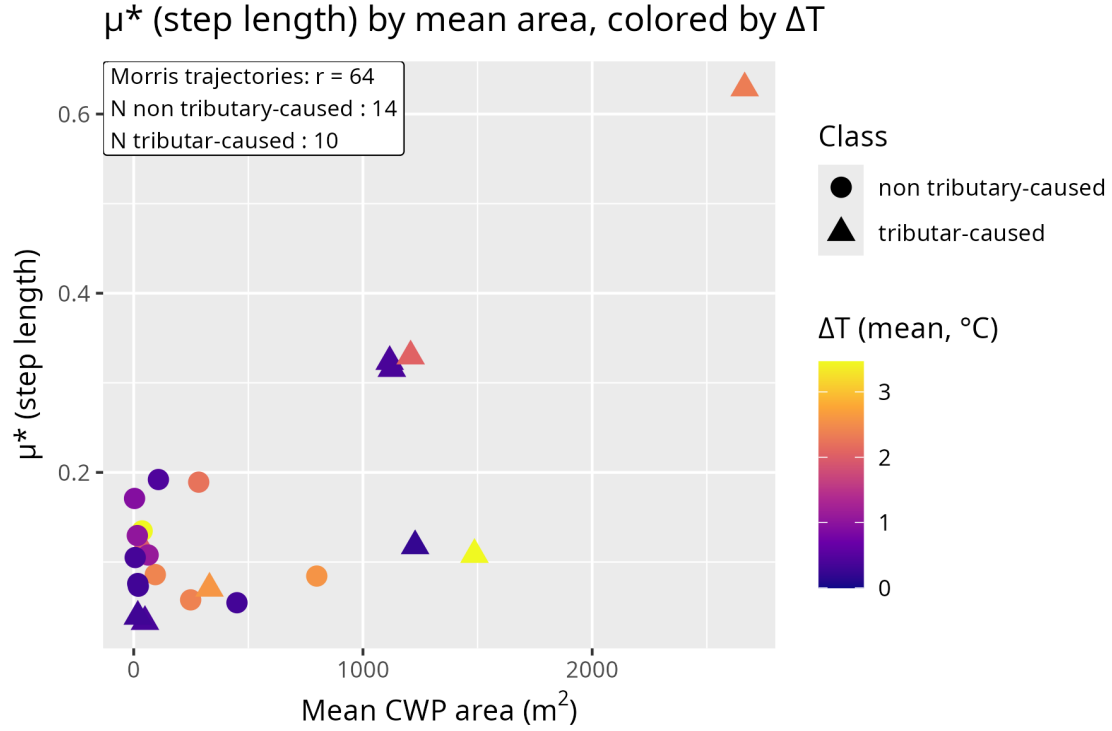


Abb. 2-5: μ^* (step length) versus mean CWP area for each of the 24 annotated locations. Color indicates the mean temperature delta to the surrounding river ($\overline{\Delta T}$, averaged across all Morris runs); shape indicates CWP class (circle: non-tributary-caused; triangle: tributary-caused).

Tab. 2-1: Regression results for μ^* (step length) predicted by mean CWP area and mean ΔT , fitted across all 24 annotated locations.

Predictor	Estimate	SE	t	p
(Intercept)	9.024×10^{-2}	3.094×10^{-2}	2.92	0.009
Mean area	1.578×10^{-4}	2.839×10^{-5}	5.56	< 0.001
Mean ΔT	-1.041×10^{-2}	1.739×10^{-2}	-0.60	0.556

Note. $R^2 = 0.611$, Adjusted $R^2 = 0.572$, $F(2, 20) = 15.73$, $p < 0.001$. Residual standard error = 0.088 on 20 degrees of freedom (one observation deleted due to missingness).

4.3 Research Question 3: Algorithm Modification

This section presents the evaluation of v3 relative to v1, covering both the quantitative Jaccard similarity comparison and the qualitative assessment of detection improvements resulting from the IDW-based reference temperature computation.

4.3.1 Output Comparison of v3 against v1

The CWP polygons produced by v3 were compared against those of v1 using the same Jaccard similarity metric applied in the output equivalence comparison of v1 and v2 (Section 3.2.2). The Jaccard similarities obtained were 0.325, 0.334, and 0.287 for the three

validation parameter combinations – substantially lower than the values of 0.936, 0.956, and 0.976 obtained for v2 versus v1. This indicates that v3 produces CWP detections that differ considerably from those of v1, which is expected: the IDW-based reference temperature computation represents a fundamental redesign of the core reference temperature step rather than a minor reimplementation, and is therefore not expected to replicate v1’s outputs closely. The relevant question is not whether v3 agrees with v1, but whether v3’s detections are ecologically more plausible.

Figure 3-6 illustrates three qualitative improvements observed in v3 relative to v1. In each panel, v1 polygons and v3 polygons are overlaid with 50% transparency, allowing areas of agreement and disagreement to be read directly from the overlap pattern.

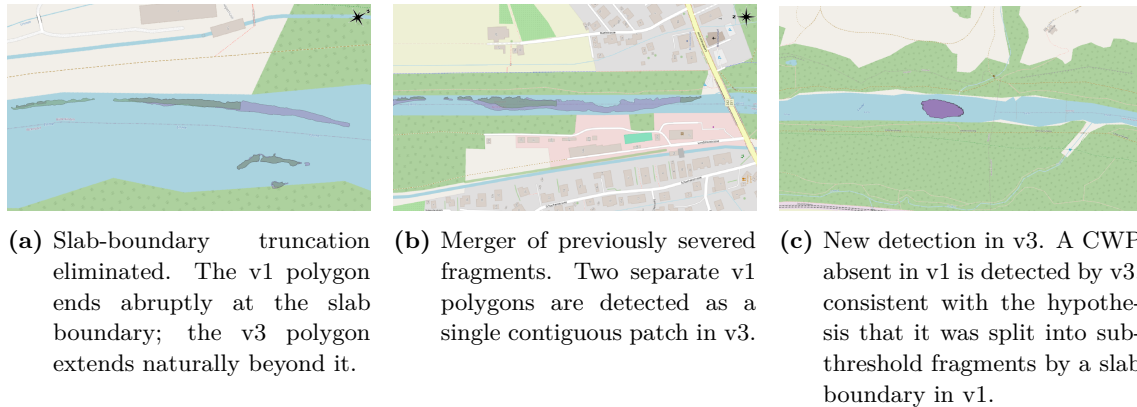


Abb. 3-6: Qualitative comparison of CWP detections between v1 and v3 at three example locations. In each panel, v1 polygons and v3 polygons are overlaid at 50% transparency; areas covered by both algorithms appear as a blend of the two colours, while areas covered by only one algorithm appear in that algorithm’s colour alone.

Panel (a) shows the most direct improvement: in v1, the detected CWP extent is truncated abruptly at a slab boundary, not because the underlying thermal anomaly ends there, but because the reference temperature changes discontinuously at that point. In v3, the spatially continuous IDW reference temperature surface removes this artificial boundary, allowing the detected patch to extend naturally beyond the former boundary location and taper off gradually as the temperature difference to the surrounding river diminishes.

Panel (b) illustrates a direct consequence of eliminating slab-boundary truncations: CWPs previously reported as two separate fragments – one on each side of a slab boundary – are now detected as a single contiguous polygon, which is likely a more accurate representation of the underlying thermal anomaly.

Panel (c) shows an example of a CWP detected by v3 that is entirely absent in v1. A likely explanation is that the corresponding thermal anomaly straddles a slab boundary in v1, causing it to be split into two sub-threshold fragments that are subsequently removed by the minimum area filter. In v3, the absence of slab boundaries allows the anomaly to be detected as a single contiguous patch above the area threshold. Whether these additional detections represent genuine CWPs missed by v1, or false positives introduced by the IDW reference temperature surface, cannot be determined without independent ground-truth validation – but the former interpretation is consistent with the mechanistic argument developed in Section 3.4.

Chapter 5

Discussion

5.1 Research Question 1: Algorithm Reimplementation

This section discusses the results of the output equivalence comparison and the runtime benchmarking between v1 and v2, and assesses whether the sensitivity analysis findings derived from v2 can be transferred to v1.

5.1.1 Difference in CWP Polygons between v1 and v2

The CWP polygons produced by v1 and v2 are in general agreement with respect to location and spatial extent, as reflected by Jaccard similarities of 0.936, 0.956, and 0.976 across the three validation runs. The observed discrepancies are confined to the pixel level – typically affecting a small number of pixels at patch boundaries – and no systematic spatial bias is apparent.

The most likely source of these discrepancies is the difference in how `terra::extract()` and `exactextractr::exact_extract()` handle partially covered pixels: as described in the Methods, the 50% coverage-fraction threshold used to approximate `terra`’s center-point inclusion rule is not exactly equivalent, and small differences in the extracted reference temperatures could propagate to pixel-level differences in the binary CWP raster. Importantly, the differences between v1 and v2 are minor and bounded in their effect: they do not introduce systematic differences in which CWPs are detected or where, but only affect a small number of boundary pixels.

The research question is specifically targeted at the behaviour of v1. The sensitivity analysis was, however, conducted on v2 as this optimized algorithm version has a substantially reduced runtime. The high Jaccard similarities, combined with the observation that the discrepancies between v1 and v2 can be attributed to a minor rounding artefact rather than to any fundamental difference in algorithmic behaviour, support the conclusion that v2 replicates the behaviour of v1 to a degree sufficient to transfer sensitivity findings. In other words, the sensitivity of CWP area to the step length, buffer pixel count, and temperature delta parameters as observed in v2 can be considered representative of the sensitivity that would have been observed had the Morris experiment been conducted on v1 directly.

5.1.2 Speed Comparison

The optimized algorithm (v2) shows substantially shorter execution times than v1 across all three tested segment lengths, with the performance advantage being most pronounced at shorter step lengths. Of particular interest is the contrasting scaling behaviour of the two implementations: while v2 maintains an approximately constant execution time of around 0.5 hours regardless of step length, v1's execution time increases as step length decreases, ranging from approximately 1.25 hours at 1200 m to approximately 2 hours at 400 m.

This difference in scaling behaviour is consistent with a mechanistic explanation rooted in how the two implementations process the river reach. v2 is loop-free: GDAL operates on the full raster in a single pass, and each pixel undergoes a fixed number of operations regardless of how many segments the river reach is divided into. The number of segments therefore has no bearing on execution time, which explains the observed constant scaling behaviour. v1, by contrast, processes the reach one slab at a time using R loops. Each loop iteration incurs both a fixed overhead – the cost of the loop machinery itself, independent of slab size – and a variable computation proportional to the raster area of that slab. At shorter step lengths, the reach is divided into a larger number of smaller slabs: the variable computation per iteration decreases while the fixed overhead remains constant, shifting the ratio of overhead-to-useful-work unfavourably and increasing total execution time. This is a plausible explanation for the observed scaling behaviour of v1, though it should be noted that this mechanistic account is a hypothesis and was not formally tested within this work.

The decision to optimize the algorithm prior to the Morris experiment was well-founded: v2 is faster than v1 across the entire parameter range tested, and the performance advantage is largest precisely at the shorter step lengths (400 m) where v1 is slowest – which happen to fall within the Morris sampling grid used in this work.

5.2 Research Question 2: Sensitivity Analysis

This section interprets the Morris sensitivity analysis results, first addressing the specific hypothesis regarding step-length sensitivity in tributary-caused versus non-tributary-caused CWP, and then discussing the overall parameter sensitivity ranking across all three varied parameters.

5.2.1 Answering Research Question 2

The boxplot comparison of μ^* values between tributary-caused and non-tributary-caused CWP shows a tendency consistent with H1: tributary-caused CWP display a wider spread in step-length sensitivity, with a higher third quartile and higher maximum values than the non-tributary-caused group. However, a closer inspection reveals that this pattern is not uniform within the tributary-caused class – some tributary-caused CWP show sensitivity values comparable to the non-tributary-caused group, while others are substantially more sensitive. This within-class heterogeneity suggests that tributary presence alone is not a sufficient condition for high sensitivity to the step length parameter, and that H1 as stated does not fully capture the situation. At the same time, the data do not support H0 either: the non-tributary-caused CWP consistently show low μ^* values, and the ele-

vated sensitivity observed in a subset of tributary-caused CWP is unlikely to be entirely coincidental. The true picture lies between the two hypotheses.

The scatter plot and linear model together point to CWP area as the more proximate driver of step-length sensitivity. Across both classes, locations with larger mean CWP area tend to show higher μ^* values, and this relationship is approximately linear. The linear model confirms this: the mean area predictor is highly significant ($p < 0.001$) with a positive slope, while the mean temperature delta predictor is not significant ($p = 0.556$), indicating no detectable independent contribution of temperature delta to step-length sensitivity after accounting for area. The color distribution in the scatter plot is consistent with this: both high and low ΔT CWP appear across the full range of μ^* values, with no clear separation by temperature.

A mechanistic explanation for the area–sensitivity relationship follows naturally from the structure of the algorithm. CWPs in a river system tend to be elongated rather than wide, since the river channel constrains lateral extent while longitudinal extent is less bounded. Large CWP are therefore predominantly long CWP. A long CWP has a higher probability of spanning the boundary between two adjacent slabs, and when this occurs, the two parts of the patch are compared against different reference temperatures – one computed from each slab’s reference strip. If those reference temperatures differ (which is more likely near a tributary inflow, where sharp longitudinal temperature gradients exist), the flagging threshold is effectively applied inconsistently across the patch, introducing variability in the detected CWP extent that is directly attributable to the step length parameter. Shorter or smaller CWP are more likely to fall entirely within a single slab, and are therefore less exposed to this effect.

Tributary inflows are then best understood not as a direct cause of step-length sensitivity, but as one mechanism that can produce large, elongated CWP – by introducing substantial volumes of cold water that generate extended thermal anomalies along the river. This explains why tributary-caused CWP tend to show higher sensitivity as a group, while also explaining why the effect is not universal: a tributary-caused CWP that is small (perhaps because the tributary discharge is low or the temperature difference modest) will not be sensitive to the step length parameter for the same reason that a small non-tributary-caused CWP is not. It also means that the data cannot speak to whether a large non-tributary-caused CWP would show comparable sensitivity to an equally large tributary-caused CWP, since no non-tributary-caused CWP larger than approximately 1000 m² is present in the dataset.

Interestingly, CWP area and mean temperature delta appear to be largely unrelated in the data, despite the intuitive expectation that a colder tributary might produce a larger thermal anomaly. This could suggest that tributary discharge plays a more important role than temperature difference in determining CWP size – a large, moderately cold tributary may produce a larger CWP than a small but very cold one. Discharge data were not collected during the survey, but could be a valuable addition in future work.

The following answer to research question 2 is therefore proposed: neither H0 nor H1 is fully supported by the data. H0 is too strong a rejection – tributary presence is associated with a tendency towards higher step-length sensitivity. However, H1 overstates the role of tributary presence as a direct determinant of sensitivity. The data suggest that the more proximate driver is CWP size: large CWP, which are disproportionately but not exclusively tributary-caused, are more susceptible to the step length parameter because they are more likely to span slab boundaries and therefore be exposed to inconsistent

reference temperatures.

Two important caveats must accompany this conclusion. First, the sample size of $n = 24$ is small, and the significant area–sensitivity relationship could reflect a chance finding. Second, and more fundamentally, the selection and classification of the 24 CWP was carried out through visual interpretation rather than algorithmically, which introduces the possibility of confirmation bias: CWP were selected on the basis of appearing variable and appearing tributary-caused or non-tributary-caused, meaning the dataset reflects the analyst’s prior expectations rather than a random or representative sample of CWP in the Emme system. For these reasons, the findings should be treated as hypothesis-generating rather than confirmatory, and replication with a larger, algorithmically selected sample would be needed before stronger conclusions can be drawn.

5.2.2 Overall Sensitivity across all Parameters

The overall μ^* distributions across all 24 annotated locations reveal a clear ranking of parameter influence with respect to detected CWP area. The buffer pixel count is consistently non-influential, with μ^* values close to zero across all locations. The corresponding σ values are similarly low, suggesting that the elementary effects of buffer pixel count are both small in magnitude and consistent across the parameter space – indicating neither a strong main effect nor meaningful interactions with the other parameters. This finding is consistent with [Tonolla and Antonetti \(2026\)](#), who report the same result in their sensitivity analysis of the CWP detection algorithm. Given this replication, the strategy of factor fixing can be applied: the buffer pixel count can be set to a fixed constant value, effectively removing it as a free parameter of the algorithm.

The temperature delta parameter shows the highest μ^* values overall, indicating that it is the most influential of the three parameters with respect to CWP area. This is also consistent with the findings of [Tonolla and Antonetti \(2026\)](#), who likewise identify temperature delta as the dominant parameter. The elevated σ values for temperature delta suggest that its elementary effects vary across the parameter space, pointing to the presence of nonlinearity or interactions with the other parameters. This is ecologically plausible: the relationship between the flagging threshold and the detected CWP area is unlikely to be linear, as small changes in temperature delta near the boundary of what constitutes a detectable thermal anomaly could produce disproportionately large changes in detected area, while changes far from this boundary may have little effect.

The step length parameter occupies an intermediate position, with μ^* values clearly above zero but consistently below those of temperature delta. Like temperature delta, step length also shows elevated σ values, suggesting interactions with the other parameters or nonlinear behaviour.

The replication of the buffer pixel and temperature delta findings from [Tonolla and Antonetti \(2026\)](#) within the present work, using an independently implemented Morris experiment on a different dataset, lends additional credibility to both sets of results and suggests that these sensitivity patterns are a robust property of the CWP detection algorithm rather than artefacts of a particular study context.

5.3 Research Question 3: Algorithm Modification

Research question 3 asked what adjustments could be made to the detection algorithm to reduce variability with respect to its parameters while also achieving plausible detection results. Based on the Morris screening findings, three modifications were implemented in v3, as described in Section 3.4: the buffer pixel count was fixed by factor fixing, the slab-based reference temperature computation was replaced by an IDW scheme removing the step length dependency, and the temperature delta parameter was retained as a user-defined input. The evaluation presented in Section 4.3 shows that v3 produces outputs that differ substantially from v1 in terms of Jaccard similarity, which is expected given the fundamental redesign of the reference temperature step. Qualitatively, however, v3 eliminates the slab-boundary cutoff artefacts identified as a direct consequence of step-length sensitivity, producing more contiguous and naturally extending CWP detections. Research question 3 can therefore be answered affirmatively: the identified parameter sensitivities can be addressed through targeted algorithmic modifications without sacrificing output plausibility.

5.4 Future improvements

The following subsections identify three methodological limitations of the present work and outline directions in which future studies could address them.

5.4.1 Subjective Selection and Classification of CWP

Both the selection and classification of the CWP were carried out subjectively based on visual inspection of the fuzzy overlay raster. This approach is prone to confirmation bias on two levels. First, the judgment about variability and the judgment about class membership were both made with the study hypothesis already in mind. CWPs that were selected are therefore likely those that were most easily attributable to one of the two classes – patches that clearly appeared variable and clearly appeared tributary-caused or non-tributary-caused. More nuanced cases, where variability was ambiguous or class attribution uncertain, were left out. The resulting dataset is consequently biased towards extreme and obvious cases, and may not be representative of the broader population of CWPs in the Emme system. Second, given that H1 of research question 2 was considered plausible prior to the analysis based on prior observations, the selection process may have been further skewed towards patches that would support H1, even if this influence was not conscious.

Two methodological improvements could reduce these limitations in future work. First, the variability assessment and CWP selection could be replaced by a fully algorithmic approach, removing human judgment from the process. Caution is however warranted: upon defining the rules of a process-based algorithm, the analyst may inadvertently encode the same biases that affect manual selection. Similarly, a data-driven approach would need to learn from a ground truth dataset that is itself produced by human labeling, meaning the bias present in the labeling process would be inherited by the model. Second, if manual classification cannot be avoided, inter-rater reliability assessment should be included: multiple independent observers, each blind to the study hypothesis, could classify the same set of candidate CWPs, and the level of agreement between raters – quantified for

example using Cohen’s κ – would provide an empirical measure of classification reliability. Such a dataset could further serve as ground truth for training a data-driven classification algorithm, combining the transparency of manual labeling with the scalability of an automated approach.

5.4.2 Sample Size

With only $n = 24$ annotated CWP locations, the dataset is small by the standards of statistical inference. While the significant area–sensitivity relationship detected by the linear model suggests that the effect is strong enough to be visible at this sample size, the risk of chance findings cannot be excluded. A larger sample would increase statistical power and reduce the probability that the observed relationships are artefacts of the specific locations selected.

5.4.3 Hidden Parameter in v3

Upon closer inspection, v3 introduces a parameter that was not explicitly treated as a free parameter during the development of the algorithm: the decay power of the inverse distance weighting scheme, which governs how quickly the influence of a reference point diminishes with distance. In the current implementation this value is fixed at 2, which is a common default, but the choice is arbitrary and its influence on the detected CWP area has not been assessed. Strictly speaking, this means that the parameter reduction achieved by v3 is incomplete: the decay power has effectively been fixed by convention rather than by a principled sensitivity analysis. A dedicated sensitivity analysis of the IDW decay power would be needed to either justify fixing it at its current value through factor fixing, or to expose it as a free parameter that the user should be able to configure.

Chapter 6

Appendix

The following generative AI systems and AI-assisted tools were utilized in the context of this work:

System Tool	/	Provider & Version	Purpose of Use
Claude Sonnet 4.6		Anthropic (2026) Version Sonnet 4.6	• Stylistic improvement of text • Spelling correction • Code debugging and coding assistance (no agentic coding) • L^AT_EX assistant (debugging and formatting) • Finding relevant literature
Claude 4.6	Opus	Anthropic (2026) Version Opus 4.6	• Complex debugging tasks • Complex writing assistance tasks such as assuring cross-chapter naming consistency

Note: The author assumes full responsibility for all generated content, its accuracy, and compliance with the standards of scientific integrity.

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